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Java SE 27 & JDK 27

Class CalendarNameProvider

java.lang.Object
    java.util.spi.LocaleServiceProvider
        java.util.spi.CalendarNameProvider

```
public abstract class CalendarNameProvider
extends LocaleServiceProvider
```

An abstract class for service providers that provide localized string representations (display names) of `Calendar` field values.

Calendar Types

Calendar types are used to specify calendar systems for which the `getDisplayName` and `getDisplayNames` methods provide calendar field value names. See `Calendar.getCalendarType()` for details.

Calendar Fields

Calendar fields are specified with the constants defined in `Calendar`. The following are calendar-common fields and their values to be supported for each calendar system.

Field	Value	Description
`Calendar.MONTH`	`Calendar.JANUARY` to `Calendar.UNDECIMBER`	Month numbering is 0-based (e.g., 0 - January, ..., 11 - December). Some calendar systems have 13 months. Month names need to be supported in both the formatting and stand-alone forms if required by the supported locales. If there's no distinction in the two forms, the same names should be returned in both of the forms.
`Calendar.DAY_OF_WEEK`	`Calendar.SUNDAY` to `Calendar.SATURDAY`	Day-of-week numbering is 1-based starting from Sunday (i.e., 1 - Sunday, ..., 7 - Saturday).
`Calendar.AM_PM`	`Calendar.AM` to `Calendar.PM`	0 - AM, 1 - PM

The following are calendar-specific fields and their values to be supported.

Calendar Type	Field	Value	Description
"gregory"	Calendar.ERA	0	GregorianCalendar.BC (BCE)
		1	GregorianCalendar.AD (CE)
"buddhist"	Calendar.ERA	0	BC (BCE)
		1	B.E. (Buddhist Era)
"japanese"	Calendar.ERA	0	Seireki (Before Meiji)
		1	Meiji
		2	Taisho
		3	Showa
		4	Heisei
		5	Reiwa
	Calendar.YEAR	1	the first year in each era. It should be returned when a long style (Calendar.LONG_FORMAT or Calendar.LONG_STANDALONE) is specified. See also the Year representation in SimpleDateFormat.
"roc"	Calendar.ERA	0	Before R.O.C.
		1	R.O.C.
"islamic"	Calendar.ERA	0	Before AH
		1	Anno Hijrah (AH)

Calendar field value names for "gregory" must be consistent with the date-time symbols provided by DateFormatSymbolsProvider.

Time zone names are supported by TimeZoneNameProvider.

Since:

1.8

See Also:

CalendarDataProvider, Locale.getUnicodeLocaleType(String)

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**CalendarNameProvider()**	Sole constructor.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
abstract String	**getDisplayName**(String calendarType, int field, int value, int style, Locale locale)	Returns the string representation (display name) of the calendar field value in the given style and locale.
abstract Map<String,Integer>	**getDisplayNames**(String calendarType, int field, int style, Locale locale)	Returns a Map containing all string representations (display names) of the Calendar field in the given style and locale and their corresponding field values.

Methods declared in class LocaleServiceProvider

getAvailableLocales, isSupportedLocale

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

CalendarNameProvider

```
protected CalendarNameProvider()
```

Sole constructor. (For invocation by subclass constructors, typically implicit.)

Method Details

getDisplayName

```
public abstract String getDisplayName(String calendarType,
                                      int field,
                                      int value,
                                      int style,
                                      Locale locale)
```

Returns the string representation (display name) of the calendar `field value` in the given `style` and `locale`. If no string representation is applicable, `null` is returned.

`field` is a `Calendar` field index, such as `Calendar.MONTH`.
The time zone fields, `Calendar.ZONE_OFFSET` and `Calendar.DST_OFFSET`, are *not* supported by this method. `null` must be returned if any time zone fields are specified.

`value` is the numeric representation of the `field` value. For example, if `field` is `Calendar.DAY_OF_WEEK`, the valid values are `Calendar.SUNDAY` to `Calendar.SATURDAY` (inclusive).

`style` gives the style of the string representation. It is one of `Calendar.SHORT_FORMAT` (`SHORT`), `Calendar.SHORT_STANDALONE`, `Calendar.LONG_FORMAT` (`LONG`), `Calendar.LONG_STANDALONE`, `Calendar.NARROW_FORMAT`, or `Calendar.NARROW_STANDALONE`.

For example, the following call will return `"Sunday"`.

```
getDisplayName("gregory", Calendar.DAY_OF_WEEK, Calendar.SUNDAY,
               Calendar.LONG_STANDALONE, Locale.ENGLISH);
```

Parameters:

`calendarType` - the calendar type. (Any calendar type given by `locale` is ignored.)

`field` - the `Calendar` field index, such as `Calendar.DAY_OF_WEEK`

`value` - the value of the `Calendar field`, such as `Calendar.MONDAY`

`style` - the string representation style: one of `Calendar.SHORT_FORMAT` (`SHORT`), `Calendar.SHORT_STANDALONE`, `Calendar.LONG_FORMAT` (`LONG`), `Calendar.LONG_STANDALONE`, `Calendar.NARROW_FORMAT`, or `Calendar.NARROW_STANDALONE`

`locale` - the desired locale

Returns:

the string representation of the `field value`, or `null` if the string representation is not applicable or the given calendar type is unknown

Throws:

`IllegalArgumentException` - if `field` or `style` is invalid

`NullPointerException` - if `locale` is `null`

See Also:

`TimeZoneNameProvider`,
`Calendar.get(int)`,
`Calendar.getDisplayName(int, int, Locale)`

getDisplayNames

```
public abstract Map<String,Integer> getDisplayNames
(String calendarType,
 int field,
 int style,
 Locale locale)
```

Returns a `Map` containing all string representations (display names) of the `Calendar field` in the given `style` and `locale` and their corresponding field values.

`field` is a `Calendar` field index, such as `Calendar.MONTH`.
The time zone fields, `Calendar.ZONE_OFFSET` and `Calendar.DST_OFFSET`, are *not* supported by this method. `null` must be returned if any time zone fields are specified.

`style` gives the style of the string representation. It must be one of `Calendar.ALL_STYLES`, `Calendar.SHORT_FORMAT` (`SHORT`), `Calendar.SHORT_STANDALONE`, `Calendar.LONG_FORMAT` (`LONG`), `Calendar.LONG_STANDALONE`, `Calendar.NARROW_FORMAT`, or `Calendar.NARROW_STANDALONE`. Note that narrow names may not be unique due to use of single characters, such as "S" for Sunday and Saturday, and that no narrow names are included in that case.

For example, the following call will return a `Map` containing `"January"` to `Calendar.JANUARY`, `"Jan"` to `Calendar.JANUARY`, `"February"` to `Calendar.FEBRUARY`, `"Feb"` to `Calendar.FEBRUARY`, and so on.

```
getDisplayNames("gregory", Calendar.MONTH, Calendar.ALL_STYLES, Locale.ENGLISH);
```

Parameters:

`calendarType` - the calendar type. (Any calendar type given by `locale` is ignored.)

`field` - the calendar field for which the display names are returned

`style` - the style applied to the display names; one of `Calendar.ALL_STYLES`, `Calendar.SHORT_FORMAT` (`SHORT`), `Calendar.SHORT_STANDALONE`, `Calendar.LONG_FORMAT` (`LONG`), `Calendar.LONG_STANDALONE`, `Calendar.NARROW_FORMAT`, or `Calendar.NARROW_STANDALONE`

`locale` - the desired locale

Returns:

a `Map` containing all display names of `field` in `style` and `locale` and their `field` values, or `null` if no display names are defined for `field`

Throws:

`NullPointerException` - if `locale` is `null`

See Also:

`Calendar.getDisplayNames(int, int, Locale)`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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